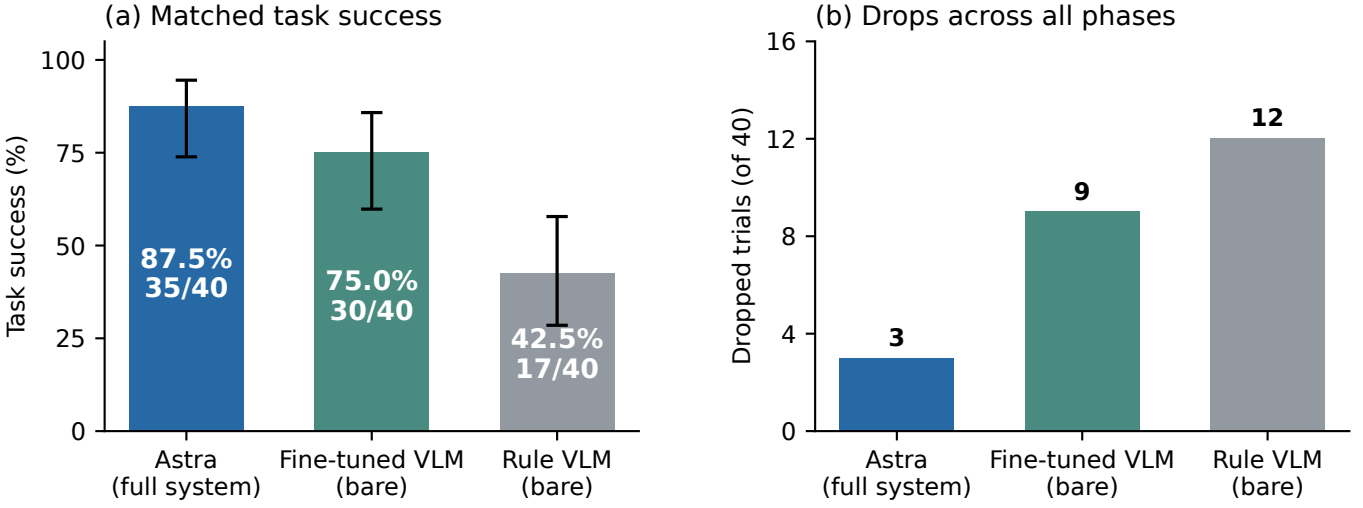

One cube, two seeds, 40 matched trials per method, no service exclusions.

Error bars: 95% Wilson intervals. Astra vs fine-tuned VLM: paired McNemar p = 0.267.

Full-system comparison. Backend and control-module effects are not isolated.